

Power Link

Definition: An arrangement of a locomotive (Loco) for working a set of coaching trains (Mail/Express/Passenger trains) from one station to another station

- **Purpose:** To manage the movement and utilization of Locomotives for passenger services
- **Preparation:** Prepared by the Mechanical Department in coordination with the Operating Department
- **Approval Authority:** CPTM (Chief Passenger Transportation Manager) is the approving authority

Points to be Kept in View while Preparing Locomotive (Power) Links

1. Preparation and Authority

- **Power/Loco links** are prepared by **Headquarters** and circulated to divisions.
- Objective: **Optimum utilization of locomotives.**

2. Timetable Verification

- Train timings should be checked against the **latest working timetable.**

3. Time of Preparation

- Links must be prepared **before commencement of the new timetable.**

4. Servicing Cycle

- Loco link must ensure that at the end of the cycle, the **locomotive returns to its nominated shed** for scheduled servicing.

5. Idling Control

- **Idling time at outstations** (waiting for next trip) should be kept **to a minimum.**

6. Fitness of Locomotives

- All locomotives included in the link must be **in good fettle (mechanical condition)** to maintain link timings and reliability.

2. Crew Link

Definition: Links drawn up for Running Staff (Crew: Driver/Assistant Loco Pilot and Guard)

- **Basis:** Crew Links are drawn up as per **HOER (Hours of Employment Regulations)**
- **Types:** Coaching links (for Mail/Express) and Freight links (part of Power Plan).
- **Scrutiny:** Vetted by the **HOER Section of the Personnel Branch**.

Points to be Kept in View while Preparing Loco Pilot's Links

1. Preparation and Verification

- Loco Pilot/Crew links are prepared by **CPRC/CTLC** and verified by the **Personnel Branch**.
- The main objective is **optimum utilization of crews**.

2. Train Timings

- Must be verified from the **latest working timetable**.

3. Time of Preparation

- Links should be prepared **before promulgation of the new timetable**.

4. Coverage of Sections

- Efforts should be made to **include all sections** in the link to avoid repeated road learning.

5. Seniority and Link Allotment

- The **link with maximum earning kilometreage** should be worked by **senior-most Loco Pilots**, and so on.

6. Special Links for Premium Trains

- Separate links should be prepared for **superfast services** (e.g., Rajdhani, Shatabdi).
- **Late running tendencies** must be considered to prevent link failures.

III. Power Plan

Definition: A document that basically indicates the **daily average number of Locos** required and planned for **Freight Services**, section-wise, for each Division

- This bare requirement of Locos for Traffic use is calculated on the basis of the traffic turn round and average number of trains run on each section
- This represents the average number of locos needed on the Division.
- **Purpose:** To assess the daily requirement of Locos and Crew on a long-term basis for a Division

RAKE LINK:

Rake link of passenger trains are issued by PCOMs of respective railways from time to time.

A **Rake Link** is a fixed operating plan that defines how each rake (set of coaches) is utilized over a period — ensuring **maximum utilization of rolling stock, adequate maintenance time, and punctual operation** of passenger services.

Every rake in passenger service is assigned a specific **link**, designed to:

- Make the best use of available rolling stock, and
- Allow sufficient time for **maintenance** at the home depot.

Purpose and Scope

- The **Rake Link Booklet** is an official publication of the **Zonal Railway**, issued for internal use.
- It details:
 - The **planned rake links** of Mail/Express and Passenger trains,
 - **Rake composition** (type and number of coaches),
 - **Marshalling order**,
 - **Permissible loads**, and
 - **Accommodation details** for each train operated by the Zone.

The rake link ensures that every rake is **optimally deployed** between trips, with adequate time allowed for **maintenance at its home depot**.

Key Characteristics

1. **Fixed Operating Pattern:**

Each rake has a **predefined cycle** covering one or multiple train services (UP & DOWN runs or shuttle trips).

2. **Optimized Maintenance Schedule:**

The link ensures the rake halts at its **home depot** at suitable intervals for cleaning, watering, mechanical inspection, and periodic maintenance.

3. **Continuity Despite Timetable Changes:**

Minor changes in train timings should **not disturb rake links**, maintaining service stability.

Types of Rake Links

1. **Long-Distance Rake Links:**

- Designed for express or mail trains operating over long distances.
- Often require **four or more rakes** depending on the total journey time and rake turnaround period.

2. **Short-Distance Rake Links:**

- Used for **local or commuter services** running multiple short-distance trips daily.
- May involve a **single rake** performing repeated shuttle runs.

Points to Consider while Preparing Rake Links

Maintenance Depot and Halting Time

- The **maintenance depot** for each rake must be identified.
- Sufficient **halt duration** should be provided at that station to enable proper maintenance and cleaning.
- Neglecting this may lead to poor coach condition (unswept carriages, non-functional lights/fans) and affect **punctuality**.

Uniformity and Public Image

- All coaches in a rake should, as far as possible, be of **same design and colour** to ensure uniform appearance and maintain a **positive public impression**.

Objectives of Rake Linking

- **Efficient utilization** of limited rolling stock.
- **Improved punctuality** by minimizing turnaround delays.
- **Systematic maintenance** scheduling.
- **Better planning** of cleaning, watering, and stocking operations.
- **Avoidance of idle rakes** and redundant coach movements.



LOCO OUTAGE AND LOCO UTILISATION

1. Definition of Loco Outage

Loco Outage represents the **average number of locomotives available for traffic use** in a 24-hour period.

It is a key performance indicator that reflects how effectively the available locomotives are being deployed for operational purposes.

$$\text{Loco Outage} = \frac{\text{Total Engine Hours for Traffic Use}}{24}$$

- The numerator represents the *total hours all locomotives were engaged in traffic use* within a day.
- Dividing by 24 (hours in a day) gives the **average number of locomotives** effectively “on line” or available to traffic.

2. Interpretation

- A higher loco outage indicates **better availability and utilization** of locomotives.
- A lower outage could imply excess idle time, maintenance backlogs, or detentions affecting loco productivity.

Outage figures can be prepared:

- **Service-wise** (Goods, Passenger, Shunting)
- **Shed-wise** (for each loco shed)
- **Division-wise** or **Railway-wise**
- **Traction-wise** (Diesel or Electric)

3. Graphical Representation (MIS Tool)

Since Diesel and Electric locomotives perform **long-distance, multi-divisional runs**, their activity can be tracked graphically over a **24-hour cycle** using a **Bar Chart** or **Line Graph**.

A **colour-coded bar chart** may represent:

Such a graphical depiction provides an effective **Management Information System (MIS)** for real-time monitoring and performance review.

4. Purpose and Utility

- Helps monitor **locomotive availability** against the sanctioned holding.
- Assists in identifying **bottlenecks** such as excessive detention at terminals, fuelling delays, or maintenance congestion.
- Enables planning for **crew link alignment, power link modification, and maintenance scheduling**.
- Aids higher management in **decision-making and resource optimization**.

5. Relation to Loco Utilisation

While **Loco Outage** measures *availability*, **Loco Utilisation** measures *productivity* — i.e., how efficiently the available locomotives are used.

$$\text{Loco Utilisation (km / day)} = \frac{\text{Total Train Kilometres Run}}{\text{Average Loco Outage}}$$

- Higher loco utilisation implies efficient use of available locos in terms of kilometres earned per day.
- Both metrics together form a vital part of the **Power Plan performance analysis**.

6. Practical Considerations

- The **target for loco outage** is fixed division-wise and reviewed periodically by the Mechanical and Operating Departments.
- The **actual outage** may vary from the target due to:
 - Bunching of trains,
 - Sudden increase in freight or passenger demand,
 - Equipment failure or maintenance delays,
 - Track blocks or operational disruptions.